

Maximilien Filion

Biomedical engineering student

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Professional Summary and Research Interests

Passionate fourth-year biomedical engineering student with a strong background in natural sciences, interdisciplinary problem-solving, leadership, and teamwork. Seeking opportunities to apply knowledge of cellular bioengineering, immunology, and microbiology to sustainable biotechnology, with the goal of addressing pressing environmental and health challenges.

Academic Qualifications

Bachelor of Applied Science, Biomedical Engineering | Current, since September 2023

University of British Columbia, GPA 84%

Diplôme d'Études Collégiales, Natural Sciences | September 2021 – May 2023

Royal Military College Saint-Jean, GPA 90%

Diplôme d'Études Secondaires, Natural Sciences | September 2015 – May 2021

Collège Charles-Lemoyne, Campus ville Sainte-Catherine, GPA 94%

Professional Experience

Officer Cadet | Current, since May 2021

Royal Military College & Canadian Forces Leadership School | Saint-Jean-sur-Richelieu, Québec

- ✚ Organizational leadership and administration
- ✚ Stress and crisis management
- ✚ Operational and tactical planning

Warrant Officer Second Class | February 2016 – May 2021

Royal Canadian Air Cadets | La Prairie, Québec

- ✚ Mentored and coached younger leaders through workshops and feedback.
- ✚ Developed and delivered instructive activities for teens daily.
- ✚ Taught teenagers safe outdoors techniques, leadership, manners and military traditions.

Awards and Honors

Governor General's Academic Collegiate Bronze Medal 2023

Usports and NAIA Academic All-Canadian 23-24

Graduating Athlete of the Year 2023

UBC Engineering Dean's List Winter 23-24, and 24-25

The Duke of Edinburgh Award – Bronze 2021

Lord Strathcona Medal 2021

Skills

<i>Wetlab</i>	<i>Drylab and Engineering</i>
✚ PCR ✚ Micro pipetting ✚ Gel Electrophoresis ✚ pH Measurement Tools ✚ Centrifuges ✚ Incubators ✚ Chromatography ✚ Spectrophotometry ✚ Basic Plasmid Amplification (E. Coli) ✚ Biosafety and waste disposal ✚ Glassware handling and cleaning	✚ C/C++/Java/Python/ Arduino ✚ MATLAB for Data Processing ✚ CAD and Rapid Prototyping ○ 3D modelling ○ Slicing Software ○ 3D Printing ✚ Soldering ✚ Woodworking and metalworking ✚ Machining ✚ Technical drawing

Educational Projects

Air Hockey Table | Software and Hardware Team Lead | January 2023 – May 2023

- ✚ Designed a system that could handle and manage the operating system, including user interface, in-game monitoring of score, and automation of hardware functionalities.
- ✚ Hardware and software test generation and debugging
- ✚ Ensured collaborative work and troubleshooted communication issues. I organized and established key focus points for group meetings, enhancing productivity.

Engineering Design Team - IGEM

Martian Biobricks | Hardware Team Member | Current, since September 2024

- ✚ Designing and testing a Caulobacter bioreactor from commercially available components.
- ✚ Working on a 3D bioprinter from commercially available components.
- ✚ Developing bioproduction pipeline of bricks made through biocementation.

Involvements

Current, since September 2023: UBC Varsity Cross-Country and Track Team

Current, since September 2023: UBC Competitive Triathlon Club

Humanitarian trip – Haiti | 2016 and 2018

Humanitarian trip – Nicaragua | 2018

Personal Interests

In my free time, I enjoy reading philosophy, exploring new ideas, and challenging my perspectives. I'm also passionate about hiking and camping, where I find both adventure and tranquility in nature.

References

References provided upon request.